

ORKDAL-THAMSHAMN, Norway

WMO: 012340

Lat: 63.3153N

Lon: 9.8494E

Elev: 13

StdP: 14.69

Time Zone: 1.00 (EUC)

Period: 06-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -1.9	(c) 3.8	(d) -5.8	(e) 4.0	(f) -1.9	(g) 0.3	(h) 5.6	(i) 4.7	(j) 25.1	(k) 33.3	(l) 22.2	(m) 34.6	(n) 9.8	(o) 210	(p) 0.682

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.0	73.5	61.4	69.5	59.6	66.5	58.7	63.4	70.0	61.4	67.2	59.6	65.0	10.3	60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
60.6	79.1	66.3	58.5	73.3	64.1	56.9	69.0	62.7	28.8	69.7	27.4	67.4	26.2	65.1	71.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 20.3	(b) 17.9	(c) 15.9	(e) -3.0	(f) 82.1	(g) 6.4	(h) 3.8	(i) -7.6	(j) 84.8	(k) -11.4	(l) 87.0	(m) -15.0	(n) 89.1	(o) -19.6	(p) 91.9		
(4) 20.3	(5) 17.9	(6) 15.9	(7) -3.3	(8) 65.7	(9) 6.6	(10) 2.8	(11) -8.1	(12) 67.7	(13) -11.9	(14) 69.3	(15) -15.7	(16) 70.9	(17) -20.4	(18) 72.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	41.7	26.1	27.9	33.7	40.2	47.9	53.7	58.8	56.9	50.9	41.6	32.3	28.8		
	DBStd	13.41	10.14	9.94	6.74	5.15	6.04	5.10	5.30	4.38	4.80	5.86	8.44	10.20		
	HDD50	3798	741	618	504	297	114	18	3	46	267	531	658			
	HDD65	8538	1206	1038	969	743	530	341	206	253	424	724	981	1123		
	CDD50	752	0	0	0	4	48	128	275	217	72	8	0	0		
	CDD65	18	0	0	0	0	0	1	14	3	0	0	0	0		
	CDH74	108	0	0	0	0	3	6	76	23	0	0	0	0		
	CDH80	17	0	0	0	0	0	0	15	2	0	0	0	0		
	WSAvg	8.0	9.0	8.4	8.6	7.9	7.8	7.8	7.5	7.1	7.5	7.7	8.2	9.1		
	PrecAvg	44.2	4.4	4.1	3.8	2.3	2.2	2.9	3.1	3.6	4.4	4.2	4.2	5.0		
(15) Precipitation	PrecMax	56.2	12.0	9.2	10.6	6.7	4.4	5.4	5.2	6.3	11.0	8.7	10.6	10.9		
	PrecMin	29.4	0.2	0.1	1.0	0.4	1.0	0.3	0.7	1.2	1.3	0.6	0.1	1.1		
	PrecStd	6.1	2.7	2.1	2.5	1.5	0.7	1.1	1.1	1.4	2.5	1.6	2.9	2.3		
	PrecStd	6.1	2.7	2.1	2.5	1.5	0.7	1.1	1.1	1.4	2.5	1.6	2.9	2.3		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	45.7	47.2	51.5	64.3	72.3	73.2	82.3	77.2	67.7	61.5	53.0	48.9		
		MCWB	41.5	41.6	43.3	51.0	58.5	58.1	65.6	62.1	56.0	51.9	48.0	42.9		
	2%	DB	43.5	44.8	47.7	55.3	65.6	69.0	75.1	70.8	63.9	57.5	48.2	46.3		
		MCWB	40.1	40.0	42.3	47.0	55.9	57.2	62.8	59.7	55.0	50.0	44.3	41.6		
	5%	DB	40.7	42.5	45.4	51.2	61.3	65.4	70.7	67.1	60.8	53.0	45.3	43.9		
		MCWB	37.8	38.6	40.5	44.5	52.6	56.4	61.9	59.5	53.8	47.5	41.9	39.8		
	10%	DB	37.8	39.7	43.1	48.5	57.6	62.1	67.2	64.3	58.4	49.9	42.1	41.1		
		MCWB	34.9	36.7	38.8	43.4	50.8	55.6	60.3	58.5	53.4	45.8	39.6	37.9		
	0.4%	WB	42.6	42.3	45.6	51.3	60.9	61.6	67.5	63.8	58.6	54.2	49.2	44.4		
		MCDB	44.6	45.7	49.2	61.1	68.5	69.2	75.7	71.1	64.1	58.1	52.3	46.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	40.5	40.5	43.4	47.9	56.5	59.3	65.0	61.9	56.6	50.9	45.1	42.3		
		MCDB	43.1	43.9	47.0	53.6	63.9	65.6	72.7	67.6	60.4	56.5	47.5	45.5		
	5%	WB	38.2	38.9	40.8	45.7	54.0	57.7	62.8	60.3	55.4	48.3	42.2	40.1		
		MCDB	40.5	41.8	44.9	49.9	60.4	63.2	68.9	65.4	58.8	52.5	44.7	43.4		
	10%	WB	35.4	36.8	39.0	43.8	51.5	55.9	60.8	58.9	54.0	46.3	39.7	37.8		
		MCDB	37.3	39.5	42.4	48.0	56.8	61.4	66.2	63.8	57.7	49.3	42.1	40.6		
(35) Mean Daily Temperature Range	5% DB	MDBR	11.1	12.0	13.0	13.6	13.6	11.9	12.0	12.5	12.2	11.4	9.6	10.6		
		MCDBR	11.0	12.8	13.9	18.4	19.9	18.5	19.0	18.3	15.5	14.9	13.1	12.8		
	5% WB	MCWBR	7.9	9.1	9.8	11.3	10.6	9.7	9.2	9.6	8.9	9.7	9.8	9.4		
		MCWBR	9.7	11.7	13.0	16.2	19.1	15.9	17.3	14.4	13.5	13.0	12.0	11.9		
	5% WB	MCWBR	7.4	8.8	9.6	10.6	10.7	9.4	9.2	8.5	8.5	9.2	9.4	9.2		
		MCWBR	7.4	8.8	9.6	10.6	10.7	9.4	9.2	8.5	8.5	9.2	9.4	9.2		
(40) Clear-Sky Solar Irradiance	taub	0.227	0.288	0.318	0.344	0.347	0.350	0.359	0.347	0.326	0.303	0.368	0.355			
	taud	2.051	2.237	2.290	2.359	2.430	2.470	2.469	2.500	2.542	2.455	2.729	2.638			
	Ebn at Noon	124	205	246	266	275	277	270	264	247	204	119	69			
	Edh at Noon	10	19	26	31	31	31	30	27	21	15	9	6			
(44) All-Sky Solar Radiation	RadAvg	49	215	620	1168	1484	1561	1468	1129	678	300	78	24			
	RadStd	8	35	84	88	86	144	136	119	75	41	14	2			

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (39 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	-257	-340	+121	N/A

Nomenclature: See separate page